

PAWAN S BANT

Aspiring AI Engineer | Applied ML, GenAI & Knowledge Graphs | M.Tech Candidate

+91-7676868328 • bantpawan@gmail.com • Bengaluru, Karnataka • GitHub

PROFESSIONAL SUMMARY

M.Tech candidate in Artificial Intelligence & Machine Learning at MS Ramaiah University of Applied Sciences (CGPA 8.39/10, graduating 2027), seeking a full-time engineering role. Builder-focused experience turning ML pipelines into evaluated, reproducible systems — including a RAG architecture with a hard constraint that locks its output and blocks the LLM from altering it, and a fraud pipeline achieving 0.9916 AUC. Multiple design choices across my projects were tested, measured, and reverted when the evidence didn't support them. After B.Tech, prepared for UPSC Civil Services before redirecting into self-directed ML training and this M.Tech.

EDUCATION

M.Tech in Artificial Intelligence & Machine Learning

2025 – 2027

MS Ramaiah University of Applied Sciences, Bengaluru | CGPA: 8.39 / 10

B.Tech in Mechanical Engineering

2015 – 2019

KLE Institute of Technology, Hubballi, Karnataka

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, Pandas, NumPy
- **Machine Learning:** Scikit-learn, Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **ML Algorithms:** Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM, K-Means, PCA
- **Generative AI & NLP:** RAG, FAISS, LLM Applications, Text Embeddings
- **Data Visualization & Deployment:** Power BI, Streamlit, Git

PROJECTS

GraphRAGXAI — Explainable Graph-RAG for Rare-Disease Diagnosis

2026

Python · Knowledge Graphs · RAG · LLMs · Neo4j · XAI | M.Tech Research Project

- Built an explainable AI diagnosis system that pairs a large medical knowledge graph — 106,658 entities linked by 1.55 million relationships — with an LLM, adding a safeguard so the AI can't silently override the computed diagnosis ranking. Correctly surfaced the right diagnosis among its top 5 suggestions 77% of the time across 70 real patient cases.
- Rigorously validated the design: tested multiple technical strategies, kept the one that measurably worked best, and dropped two alternatives that performed worse in testing. The system's stated reasoning matched its actual decision across all 350 test cases (100% consistent).

Credit Card Fraud Detection System

2025

Python · XGBoost · LightGBM · SHAP · Risk Analytics · MLOps

- Built end-to-end ML pipeline on a 1M+ transaction Kaggle dataset with 110 engineered features and 55% dimensionality reduction, achieving 0.9916 AUC, 0.9083 AP, and 89% recall via a production-grade XGBoost pipeline with SMOTE-based imbalance handling and SHAP interpretability.
- Designed an MLOps-ready modular architecture with reproducible experiments and deployable preprocessing; EDA uncovered email/device and weekend/night fraud patterns, and targeted missing-value handling cut fraud-detection errors by 8.1%.

Job Automation Risk Prediction — Interpretable Ensemble ML

2025

Python · AdaBoost · SHAP · Hybrid Feature Selection | M.Tech Research Project

- Built a supervised regression pipeline on a 3,000-record synthetic labor-market dataset, using hybrid feature selection across seven techniques (correlation, RF/GB importance, permutation, RFE) and tuning an AdaBoost ensemble via 10-fold cross-validated RandomizedSearchCV to reach R^2 of 0.871 — the strongest of seven models compared, including XGBoost, Random Forest, and MLP.
- Applied SHAP analysis showing job title and risk category as the dominant predictors; salary and education showed no statistically significant relationship to automation risk (ANOVA $p=0.134$; $\chi^2 p=0.843$).

PROFESSIONAL DEVELOPMENT

Data Science Trainee — Besant Technologies

2023 – 2024

- Hands-on training in ML algorithms, Power BI, and Azure ML deployment across three end-to-end projects.